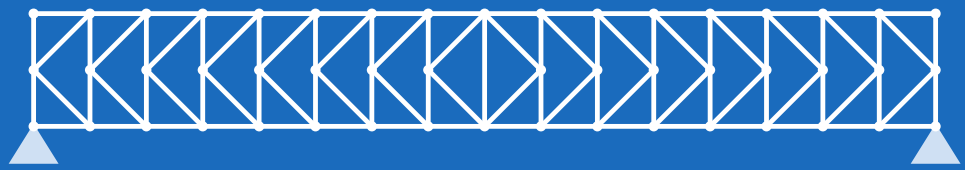
TRUSS REPORT

K-truss bridge 64 m (rigid joints)

Plane truss — analysis and CIRSOC 301 rod checks



Author A. Student · Course Structures I · 13/10/2026

MAX TENSION	MAX COMPRESSION	DEFLECTION	MOST USED ROD	BUCKLING Λ_1	WEIGHT
+943.8 kN	-944.0 kN	84.0 mm	0.75	8.66	15679 kg
rod 8	rod 24	L/762 · limit L/300	rod 8 · H1-1a, axial + bending	the load can grow ×8.7	carries 900.0 kN · 6× own weight

The model

Span	64.00 m
Height	8.00 m
Nodes / rods	50 / 97
Supports	pin at 0, rollerX at 16
Loads	15 nodal, total 900.0 kN
Rod families	5, with their sections:
Bottom chord	HEA 220, F24 (CIRSOC) / A36
Diagonal	SHS 150x150x6, F24 (CIRSOC) / A36
End post	SHS 200x200x8, F24 (CIRSOC) / A36
Top chord	HEB 220, F24 (CIRSOC) / A36
Vertical	SHS 120x120x5, F24 (CIRSOC) / A36

What the analysis found

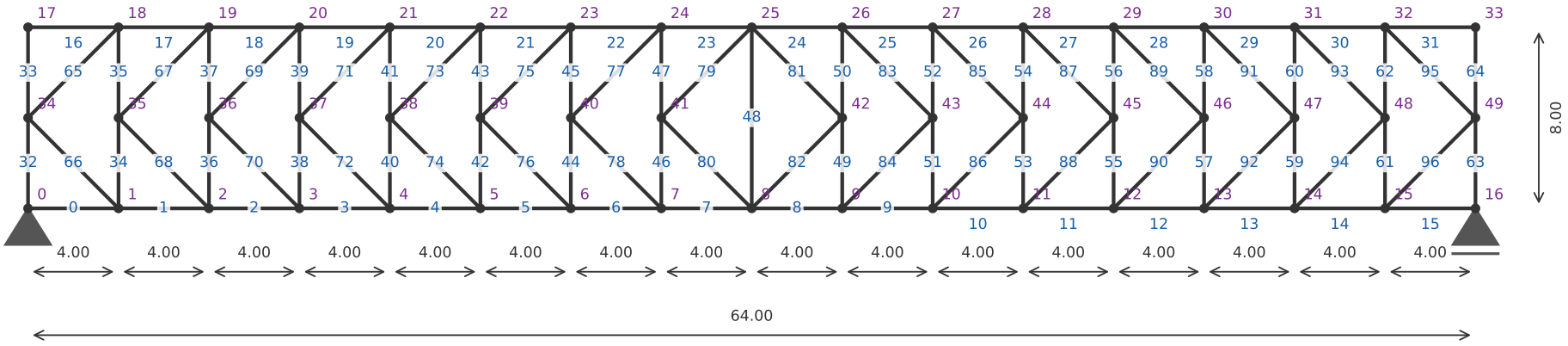
Stability	Statically indeterminate to degree 144 (deformations + r − DOF = 291 + 3 − 150 = 144): the forces also depend on the stiffness of the rods.
Most used rod	rod 8 at 0.75 (H1-1a, axial + bending)
Rods over capacity	0 of 97 checked
Buckling	$\lambda_1 = 8.66$
Shear and moment	present — see the Shear, Moment, Joints and Rod detail sheets
Joint moment	largest -4.94 kN·m at joint 0 (clockwise, from rod 0)

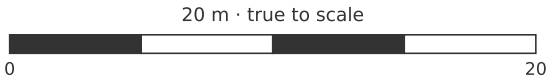
Contents

2 Geometry
3 Axial forces
4 Deflection and stress
5 Reactions
6 Utilisation
7 Buckling
8 Shear V
9 Moment M
10 Joints · 3 sheets
13 Rods · 2 sheets
15 Rod detail · 6 sheets
21 Explain rod 8 and 7

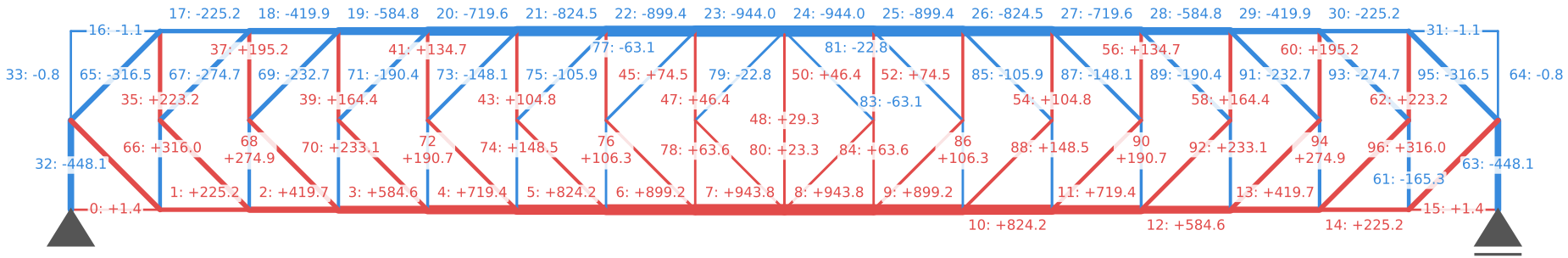
Results are an aid to a qualified professional, who must check them. Rod checks to CIRSOC 301 as implemented; first-order analysis; joints and connections are not checked here.

Node numbers in purple, rod numbers in blue. Dimensions in m.





Axial force N (kN): red pulls (+), blue pushes (−), grey carries nothing; line width follows |N| (largest 944.0).

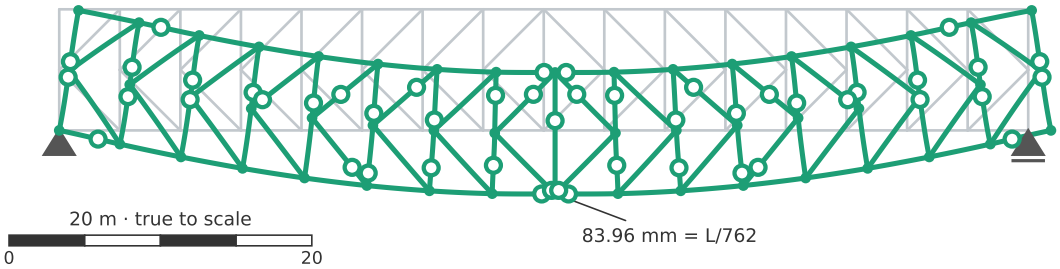


20 m · true to scale

0 20

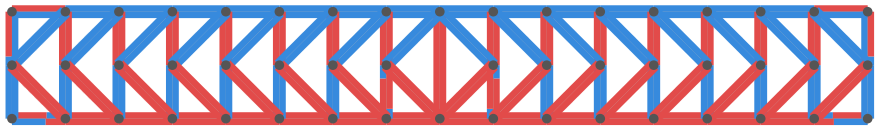
14 of 97 labels left out where they would overlap; every force is in the Rods table.

Elastic line: the deflected shape (green) over the truss (grey), every rigid rod drawn as the curve it bends into. Geometry to scale; displacements magnified × 50. ○ = point of contraflexure (M = 0), where a rod changes the way it bends: 49 of them. Largest vertical deflection 83.96 mm against the limit L/300 = 213.3 mm: OK.

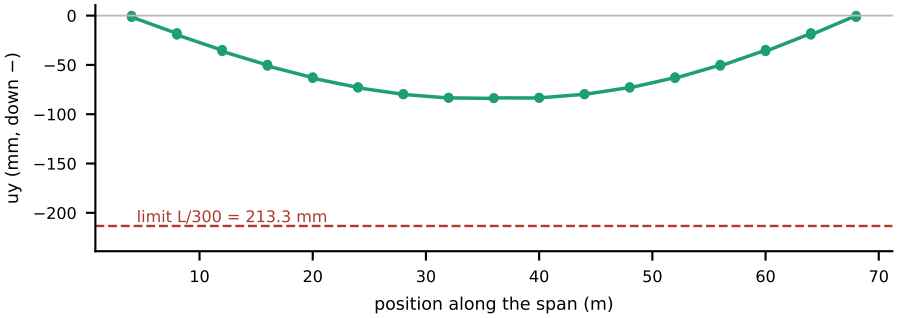


Fibre stress top and bottom face of each member

red = tension, blue = compression. Axial force and bending together (N/A ± M·c/I); a band flips colour at a point of contraflexure.

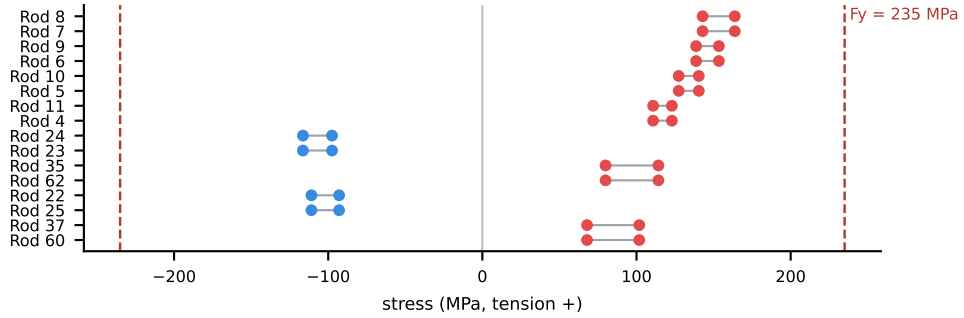


Deflection along the span vertical movement of every joint



Largest 83.96 mm at joint 8 = L/762. Within the limit.

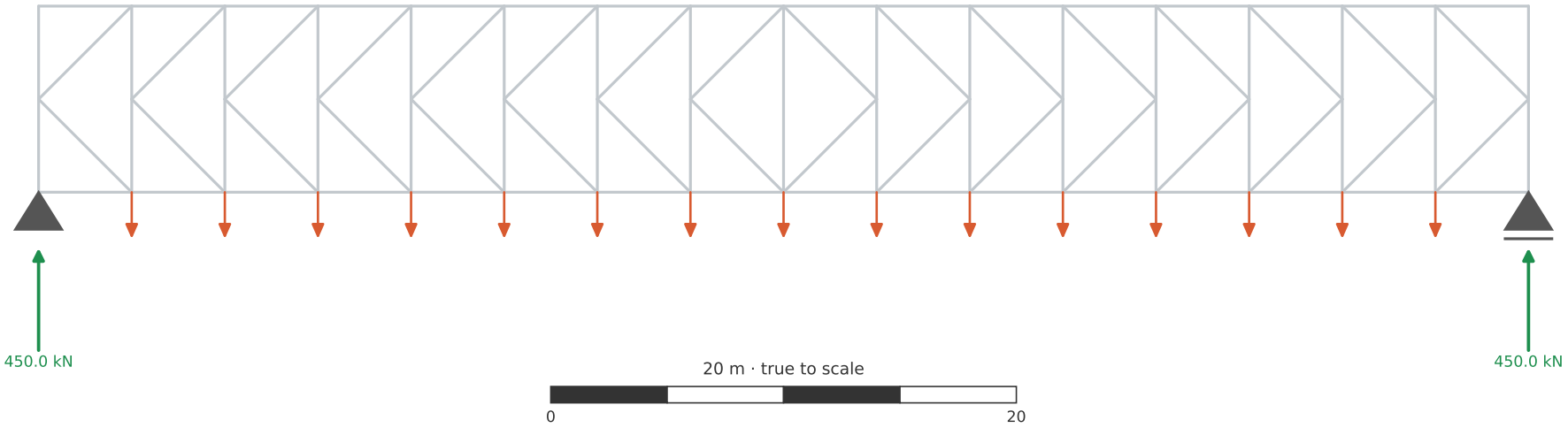
Fibre stress values the two faces of each rod, MPa



The 16 most stressed rods; the bar between a rod's two dots is its bending.

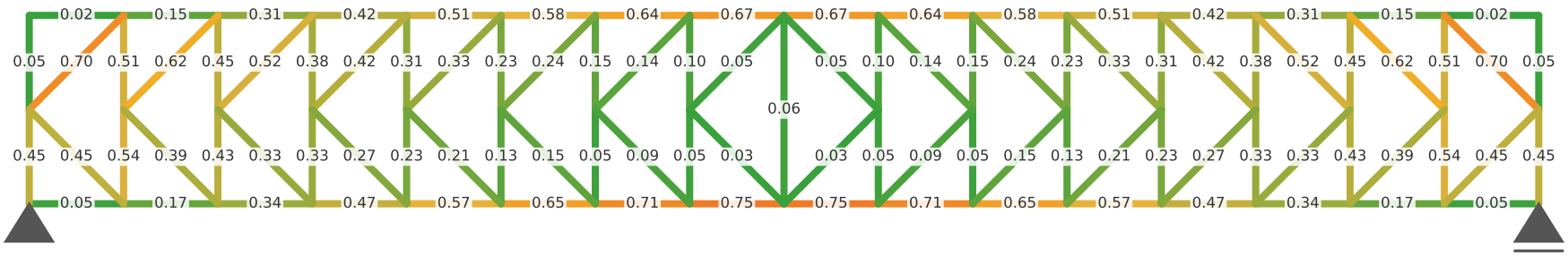
Support reactions (green arrows), as the supports push on the truss, against the loads (orange); Ry positive upwards. The sums balance the loads.

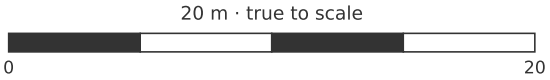
15 loads of 60.0 kN each, hung from the joints

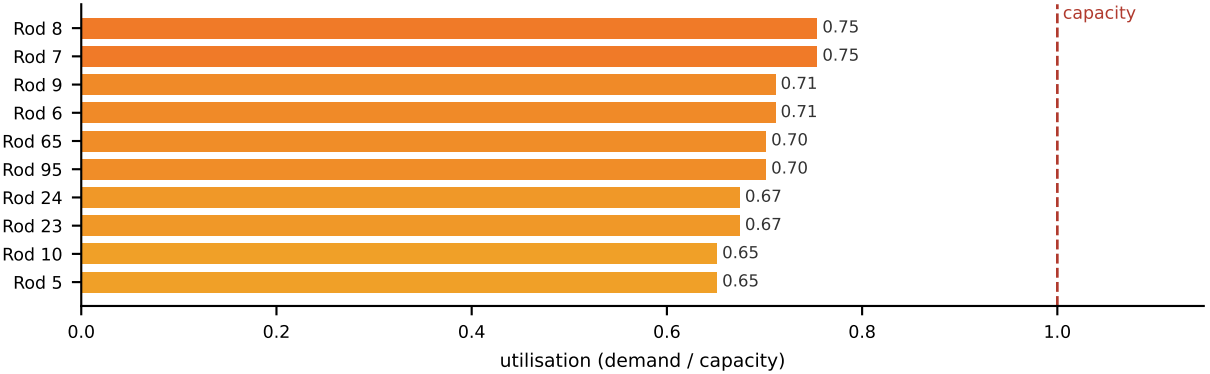


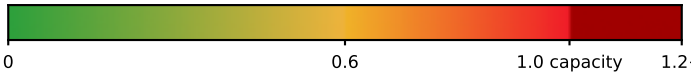
Node	Type	Rx (kN)	Ry (kN)
0	pin	0.00	450.00
16	rollerX	0.00	450.00
Σ		0.00	900.00

Utilisation: demand over CIRSOC 301 capacity. Most used rod 8 at 0.75 (H1-1a, axial + bending). Every checked rod is within capacity.









grey = no steel section, not checked

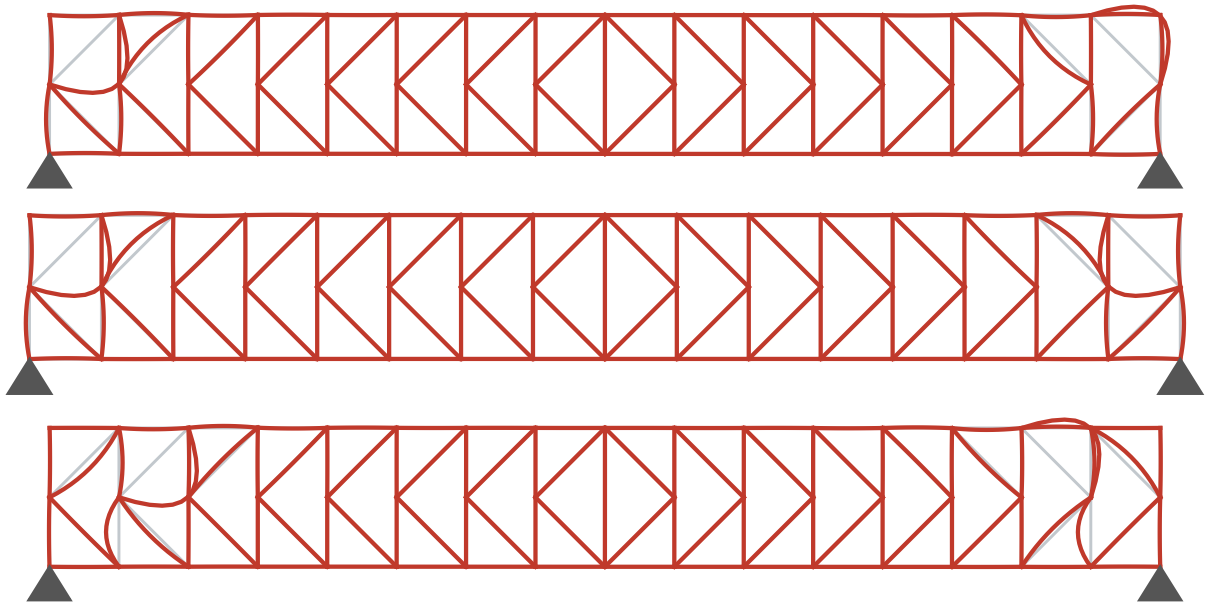
The 10 most used of 97 checked rods; the rest are in the Rods table.

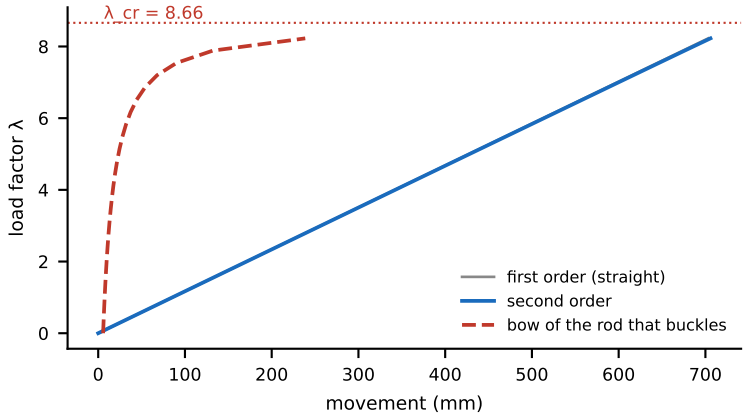
λ is the factor on the load as it stands. $\lambda_1 = 8.66$, so the load can grow 8.7× before it buckles. 2 modes share that λ: that many equal members buckle at the same load, so stiffening one of them buys nothing.

Mode 1: λ = 8.66
rod 95 bows

Mode 2: λ = 8.66
rod 65 bows

Mode 3: λ = 9.00
rod 67 bows

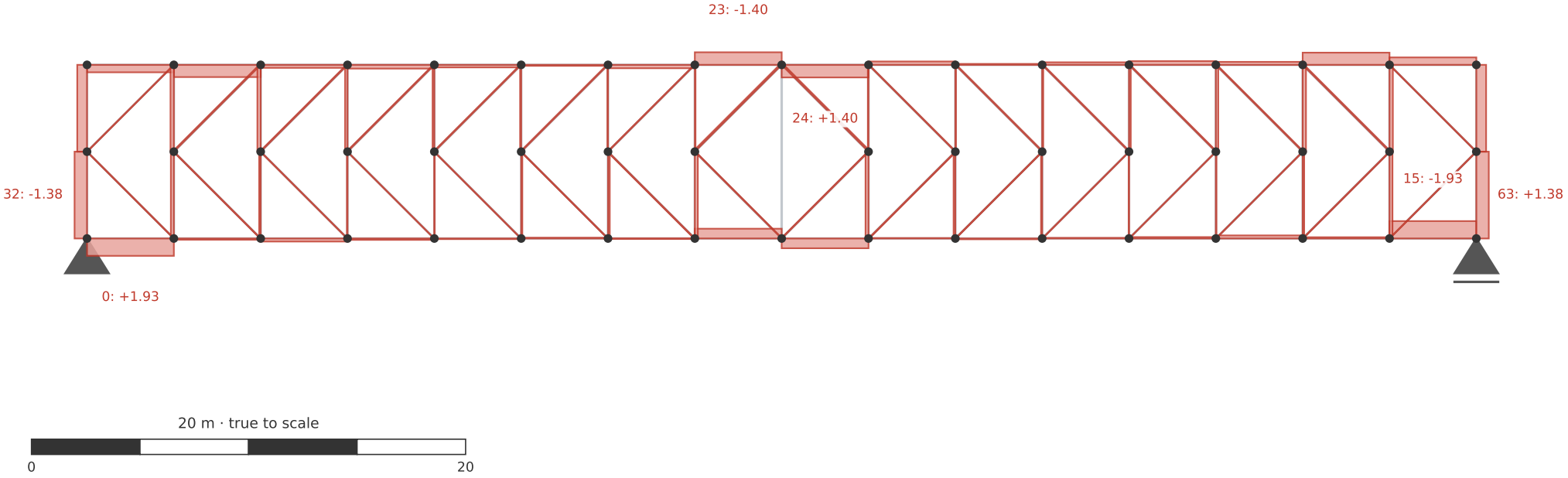




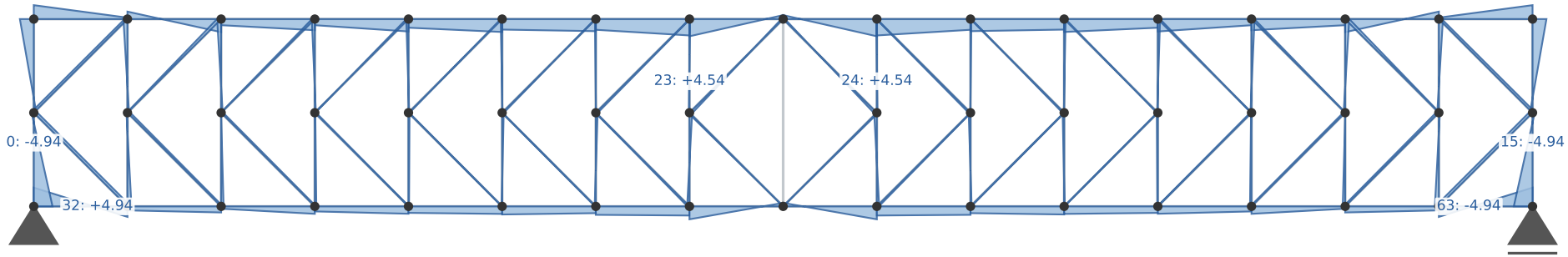
The second-order line bends away from the first-order one and runs off as λ nears λ_cr: that gap IS the amplification every second-order design rule is about. Started from an initial bow of L/1000 = 5.66 mm in the first mode's shape.

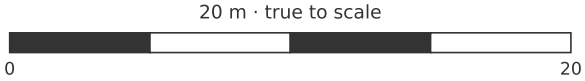
Elastic buckling of the IDEAL truss, so every λ is an UPPER bound: imperfections, residual stresses and yielding are not modelled, and each rod is cut in two so it can bow between its joints. Buckling OUT of the plane is the rod check's job (CIRSOC 301 E3, minor radius) and is in the Utilisation and Rods sheets, not here. A mode has a shape, not a size: the shapes are drawn exaggerated.

Shear force V along every rigid rod. Biggest +1.93 kN in rod 0; the band is drawn off each member, proportional to the value, and labelled on the six largest.

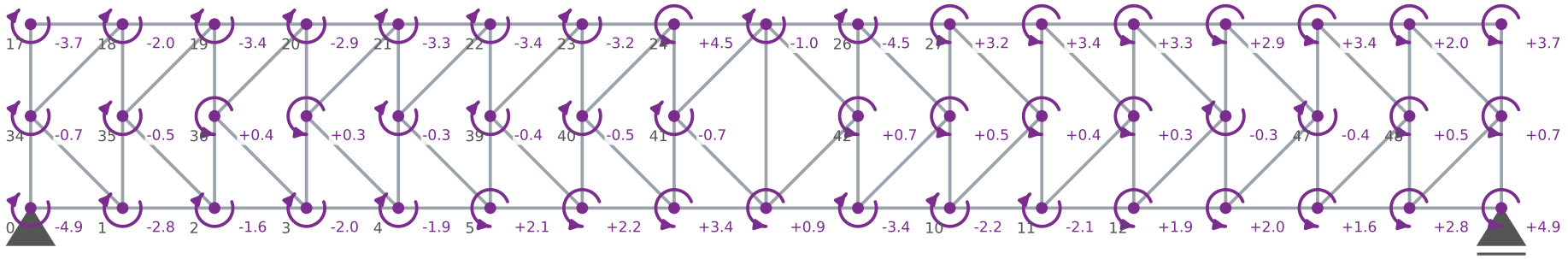


Bending moment M along every rigid rod, drawn on the face it puts in TENSION, as on screen (sagging positive: below a horizontal rod). Biggest -4.94 kN·m in rod 0; the band is drawn off each member, proportional to the value, and labelled on the six largest.





The moment at each joint: the largest one any single rod hands it -- what a rigid connection is designed for -- with its sign, which way it turns the joint (the arrow, anticlockwise = +, as on screen), and which face of that rod it puts in tension. Σ M is the check: ~0 at a free joint by equilibrium, the support's own moment at a fixed one.



20 m · true to scale

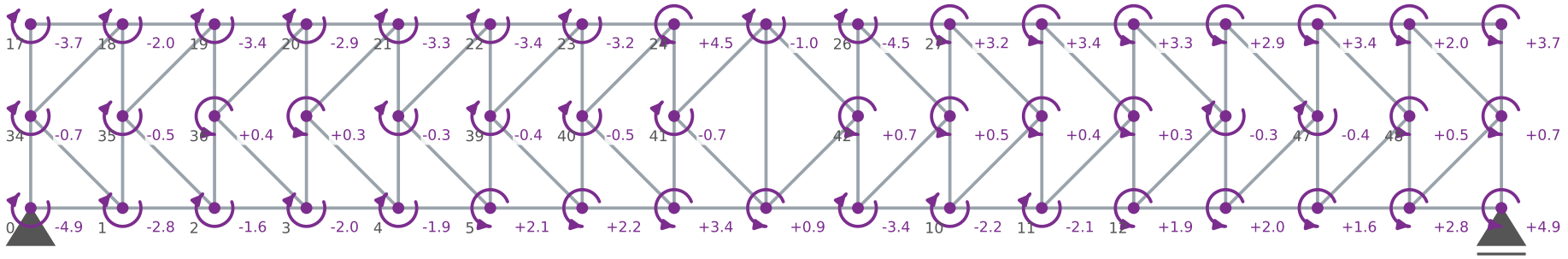
0 20

22 of 100 labels left out where they would overlap; every joint moment is in the Joints table.

Joint	Type	Max |M| (kN·m)	±	Turns it	Rod	Tension	Σ M check
0	rigid · pin	4.94	−	clockwise	0	top	0.00 ✓
16	rigid · rollerX	4.94	+	anticlockwise	15	top	0.00 ✓
26	rigid	4.54	−	clockwise	24	bottom	0.00 ✓
24	rigid	4.54	+	anticlockwise	23	bottom	0.00 ✓
33	rigid	3.67	+	anticlockwise	31	top	0.00 ✓
17	rigid	3.67	−	clockwise	16	top	0.00 ✓
31	rigid	3.45	+	anticlockwise	30	bottom	0.00 ✓
19	rigid	3.45	−	clockwise	17	bottom	0.00 ✓
9	rigid	3.41	−	clockwise	8	bottom	0.00 ✓
7	rigid	3.41	+	anticlockwise	7	bottom	0.00 ✓
28	rigid	3.40	+	anticlockwise	27	bottom	0.00 ✓

Joint	Type	Max |M| (kN·m)	±	Turns it	Rod	Tension	Σ M check
22	rigid	3.40	−	clockwise	20	bottom	0.00 ✓
21	rigid	3.27	−	clockwise	19	bottom	0.00 ✓
29	rigid	3.27	+	anticlockwise	28	bottom	0.00 ✓
27	rigid	3.19	+	anticlockwise	26	bottom	0.00 ✓
23	rigid	3.19	−	clockwise	21	bottom	0.00 ✓
20	rigid	2.94	−	clockwise	18	bottom	0.00 ✓
30	rigid	2.94	+	anticlockwise	29	bottom	0.00 ✓
1	rigid	2.77	−	clockwise	0	bottom	0.00 ✓
15	rigid	2.77	+	anticlockwise	15	bottom	0.00 ✓
6	rigid	2.23	+	anticlockwise	6	bottom	0.00 ✓
10	rigid	2.23	−	clockwise	9	bottom	0.00 ✓

The moment at each joint: the largest one any single rod hands it -- what a rigid connection is designed for -- with its sign, which way it turns the joint (the arrow, anticlockwise = +, as on screen), and which face of that rod it puts in tension. Σ M is the check: ~0 at a free joint by equilibrium, the support's own moment at a fixed one.



20 m · true to scale

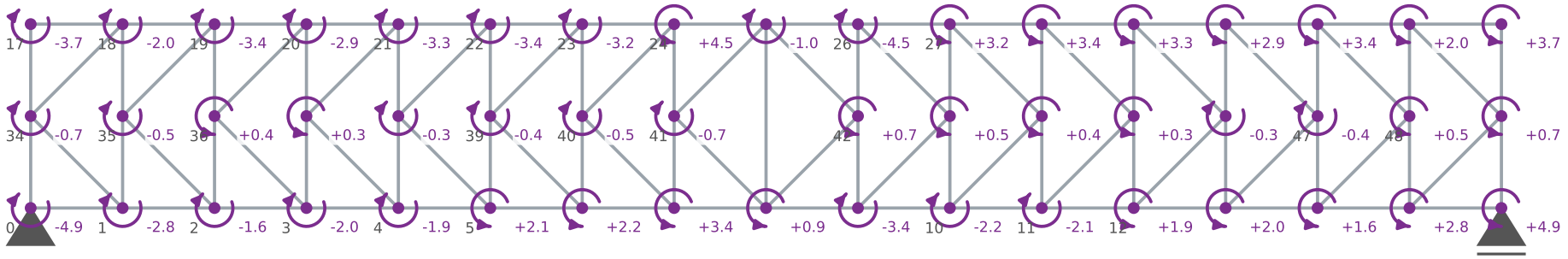
0 20

22 of 100 labels left out where they would overlap; every joint moment is in the Joints table.

Joint	Type	Max |M| (kN·m)	±	Turns it	Rod	Tension	Σ M check
11	rigid	2.14	−	clockwise	10	bottom	0.00 ✓
5	rigid	2.14	+	anticlockwise	5	bottom	0.00 ✓
32	rigid	1.97	+	anticlockwise	30	top	0.00 ✓
18	rigid	1.97	−	clockwise	17	top	0.00 ✓
3	rigid	1.97	−	clockwise	2	bottom	0.00 ✓
13	rigid	1.97	+	anticlockwise	13	bottom	0.00 ✓
12	rigid	1.94	+	anticlockwise	12	bottom	0.00 ✓
4	rigid	1.94	−	clockwise	3	bottom	0.00 ✓
14	rigid	1.60	+	anticlockwise	14	bottom	0.00 ✓
2	rigid	1.60	−	clockwise	1	bottom	0.00 ✓
25	rigid	1.04	−	clockwise	24	top	0.00 ✓

Joint	Type	Max |M| (kN·m)	±	Turns it	Rod	Tension	Σ M check
8	rigid	0.93	+	anticlockwise	7	top	0.00 ✓
42	rigid	0.70	+	anticlockwise	49	left	0.00 ✓
41	rigid	0.70	−	clockwise	46	right	0.00 ✓
34	rigid	0.67	−	clockwise	66	top	0.00 ✓
49	rigid	0.67	+	anticlockwise	96	top	0.00 ✓
48	rigid	0.53	+	anticlockwise	94	top	0.00 ✓
35	rigid	0.53	−	clockwise	68	top	0.00 ✓
43	rigid	0.48	+	anticlockwise	51	left	0.00 ✓
40	rigid	0.48	−	clockwise	44	right	0.00 ✓
36	rigid	0.41	+	anticlockwise	37	right	0.00 ✓
47	rigid	0.41	−	clockwise	60	left	0.00 ✓

The moment at each joint: the largest one any single rod hands it -- what a rigid connection is designed for -- with its sign, which way it turns the joint (the arrow, anticlockwise = +, as on screen), and which face of that rod it puts in tension. Σ M is the check: ~0 at a free joint by equilibrium, the support's own moment at a fixed one.



20 m · true to scale

0 20

22 of 100 labels left out where they would overlap; every joint moment is in the Joints table.

Joint	Type	Max |M| (kN·m)	±	Turns it	Rod	Tension	Σ M check
39	rigid	0.40	−	clockwise	42	right	0.00 ✓
44	rigid	0.40	+	anticlockwise	53	left	0.00 ✓
46	rigid	0.28	−	clockwise	58	left	0.00 ✓

Joint	Type	Max |M| (kN·m)	±	Turns it	Rod	Tension	Σ M check
37	rigid	0.28	+	anticlockwise	39	right	0.00 ✓
38	rigid	0.27	−	clockwise	40	right	0.00 ✓
45	rigid	0.27	+	anticlockwise	55	left	0.00 ✓

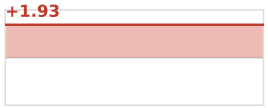
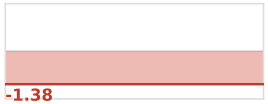
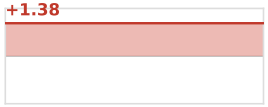
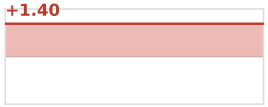
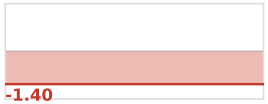
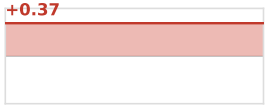
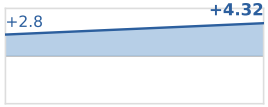
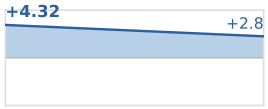
Every rod: its end nodes, family and steel section, length, axial force N (+ pulls, − pushes), the CIRSOC 301 check that governs it and how much of its capacity that uses (Use = demand / capacity; OVER above 1.0).

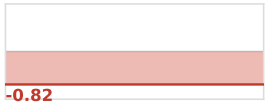
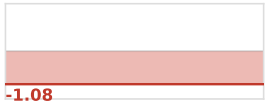
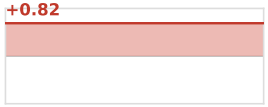
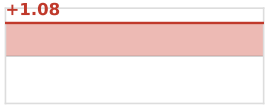
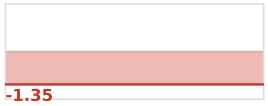
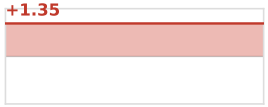
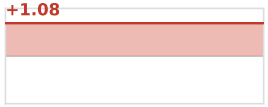
Rod	Nodes	Family	Section	L (m)	N (kN)	Governing check	Use
0	0–1	Bottom chord	HEA 220	4.00	+1.38	H1-1b, axial + bending	0.05 OK
1	1–2	Bottom chord	HEA 220	4.00	+225.18	normal stress (H.3.4)	0.17 OK
2	2–3	Bottom chord	HEA 220	4.00	+419.68	H1-1a, axial + bending	0.34 OK
3	3–4	Bottom chord	HEA 220	4.00	+584.55	H1-1a, axial + bending	0.47 OK
4	4–5	Bottom chord	HEA 220	4.00	+719.38	H1-1a, axial + bending	0.57 OK
5	5–6	Bottom chord	HEA 220	4.00	+824.23	H1-1a, axial + bending	0.65 OK
6	6–7	Bottom chord	HEA 220	4.00	+899.18	H1-1a, axial + bending	0.71 OK
7	7–8	Bottom chord	HEA 220	4.00	+943.77	H1-1a, axial + bending	0.75 OK
8	8–9	Bottom chord	HEA 220	4.00	+943.77	H1-1a, axial + bending	0.75 OK
9	9–10	Bottom chord	HEA 220	4.00	+899.18	H1-1a, axial + bending	0.71 OK
10	10–11	Bottom chord	HEA 220	4.00	+824.23	H1-1a, axial + bending	0.65 OK
11	11–12	Bottom chord	HEA 220	4.00	+719.38	H1-1a, axial + bending	0.57 OK
12	12–13	Bottom chord	HEA 220	4.00	+584.55	H1-1a, axial + bending	0.47 OK
13	13–14	Bottom chord	HEA 220	4.00	+419.68	H1-1a, axial + bending	0.34 OK
14	14–15	Bottom chord	HEA 220	4.00	+225.18	normal stress (H.3.4)	0.17 OK
15	15–16	Bottom chord	HEA 220	4.00	+1.38	H1-1b, axial + bending	0.05 OK
16	17–18	Top chord	HEB 220	4.00	-1.08	H1-1b, axial + bending	0.02 OK
17	18–19	Top chord	HEB 220	4.00	-225.22	inelastic buckling (E3-2)	0.15 OK
18	19–20	Top chord	HEB 220	4.00	-419.88	H1-1a, axial + bending	0.31 OK
19	20–21	Top chord	HEB 220	4.00	-584.75	H1-1a, axial + bending	0.42 OK
20	21–22	Top chord	HEB 220	4.00	-719.61	H1-1a, axial + bending	0.51 OK
21	22–23	Top chord	HEB 220	4.00	-824.47	H1-1a, axial + bending	0.58 OK
22	23–24	Top chord	HEB 220	4.00	-899.44	H1-1a, axial + bending	0.64 OK
23	24–25	Top chord	HEB 220	4.00	-944.03	H1-1a, axial + bending	0.67 OK
24	25–26	Top chord	HEB 220	4.00	-944.03	H1-1a, axial + bending	0.67 OK
25	26–27	Top chord	HEB 220	4.00	-899.44	H1-1a, axial + bending	0.64 OK
26	27–28	Top chord	HEB 220	4.00	-824.47	H1-1a, axial + bending	0.58 OK
27	28–29	Top chord	HEB 220	4.00	-719.61	H1-1a, axial + bending	0.51 OK
28	29–30	Top chord	HEB 220	4.00	-584.75	H1-1a, axial + bending	0.42 OK
29	30–31	Top chord	HEB 220	4.00	-419.88	H1-1a, axial + bending	0.31 OK
30	31–32	Top chord	HEB 220	4.00	-225.22	inelastic buckling (E3-2)	0.15 OK
31	32–33	Top chord	HEB 220	4.00	-1.08	H1-1b, axial + bending	0.02 OK
32	0–34	End post	SHS 200x200x8	4.00	-448.07	H1-1a, axial + bending	0.45 OK
33	34–17	End post	SHS 200x200x8	4.00	-0.82	H1-1b, axial + bending	0.05 OK
34	1–35	Vertical	SHS 120x120x5	4.00	-165.26	H1-1a, axial + bending	0.54 OK
35	35–18	Vertical	SHS 120x120x5	4.00	+223.25	H1-1a, axial + bending	0.51 OK
36	2–36	Vertical	SHS 120x120x5	4.00	-134.18	H1-1a, axial + bending	0.43 OK
37	36–19	Vertical	SHS 120x120x5	4.00	+195.15	H1-1a, axial + bending	0.45 OK
38	3–37	Vertical	SHS 120x120x5	4.00	-105.02	H1-1a, axial + bending	0.33 OK
39	37–20	Vertical	SHS 120x120x5	4.00	+164.41	H1-1a, axial + bending	0.38 OK
40	4–38	Vertical	SHS 120x120x5	4.00	-74.98	H1-1a, axial + bending	0.23 OK
41	38–21	Vertical	SHS 120x120x5	4.00	+134.67	H1-1a, axial + bending	0.31 OK
42	5–39	Vertical	SHS 120x120x5	4.00	-45.17	inelastic buckling (E3-2)	0.13 OK
43	39–22	Vertical	SHS 120x120x5	4.00	+104.81	H1-1a, axial + bending	0.23 OK
44	6–40	Vertical	SHS 120x120x5	4.00	-15.06	H1-1b, axial + bending	0.05 OK
45	40–23	Vertical	SHS 120x120x5	4.00	+74.52	normal stress (H.3.4)	0.15 OK
46	7–41	Vertical	SHS 120x120x5	4.00	+13.88	H1-1b, axial + bending	0.05 OK
47	41–24	Vertical	SHS 120x120x5	4.00	+46.36	normal stress (H.3.4)	0.10 OK
48	8–25	Vertical	SHS 120x120x5	8.00	+29.27	normal stress (H.3.4)	0.06 OK
49	9–42	Vertical	SHS 120x120x5	4.00	+13.88	H1-1b, axial + bending	0.05 OK
50	42–26	Vertical	SHS 120x120x5	4.00	+46.36	normal stress (H.3.4)	0.10 OK
51	10–43	Vertical	SHS 120x120x5	4.00	-15.06	H1-1b, axial + bending	0.05 OK

Every rod: its end nodes, family and steel section, length, axial force N (+ pulls, − pushes), the CIRSOC 301 check that governs it and how much of its capacity that uses (Use = demand / capacity; OVER above 1.0).

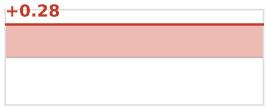
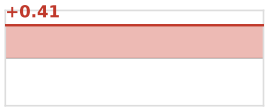
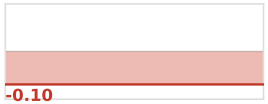
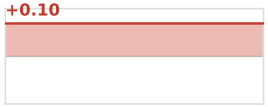
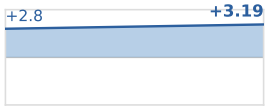
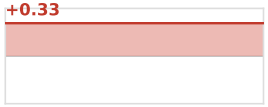
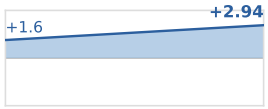
Rod	Nodes	Family	Section	L (m)	N (kN)	Governing check	Use
52	43–27	Vertical	SHS 120x120x5	4.00	+74.52	normal stress (H.3.4)	0.15 OK
53	11–44	Vertical	SHS 120x120x5	4.00	-45.17	inelastic buckling (E3-2)	0.13 OK
54	44–28	Vertical	SHS 120x120x5	4.00	+104.81	H1-1a, axial + bending	0.23 OK
55	12–45	Vertical	SHS 120x120x5	4.00	-74.98	H1-1a, axial + bending	0.23 OK
56	45–29	Vertical	SHS 120x120x5	4.00	+134.67	H1-1a, axial + bending	0.31 OK
57	13–46	Vertical	SHS 120x120x5	4.00	-105.02	H1-1a, axial + bending	0.33 OK
58	46–30	Vertical	SHS 120x120x5	4.00	+164.41	H1-1a, axial + bending	0.38 OK
59	14–47	Vertical	SHS 120x120x5	4.00	-134.18	H1-1a, axial + bending	0.43 OK
60	47–31	Vertical	SHS 120x120x5	4.00	+195.15	H1-1a, axial + bending	0.45 OK
61	15–48	Vertical	SHS 120x120x5	4.00	-165.26	H1-1a, axial + bending	0.54 OK
62	48–32	Vertical	SHS 120x120x5	4.00	+223.25	H1-1a, axial + bending	0.51 OK
63	16–49	End post	SHS 200x200x8	4.00	-448.07	H1-1a, axial + bending	0.45 OK
64	49–33	End post	SHS 200x200x8	4.00	-0.82	H1-1b, axial + bending	0.05 OK
65	34–18	Diagonal	SHS 150x150x6	5.66	-316.46	H1-1a, axial + bending	0.70 OK
66	34–1	Diagonal	SHS 150x150x6	5.66	+316.03	H1-1a, axial + bending	0.45 OK
67	35–19	Diagonal	SHS 150x150x6	5.66	-274.65	H1-1a, axial + bending	0.62 OK
68	35–2	Diagonal	SHS 150x150x6	5.66	+274.86	H1-1a, axial + bending	0.39 OK
69	36–20	Diagonal	SHS 150x150x6	5.66	-232.71	H1-1a, axial + bending	0.52 OK
70	36–3	Diagonal	SHS 150x150x6	5.66	+233.09	H1-1a, axial + bending	0.33 OK
71	37–21	Diagonal	SHS 150x150x6	5.66	-190.35	H1-1a, axial + bending	0.42 OK
72	37–4	Diagonal	SHS 150x150x6	5.66	+190.74	H1-1a, axial + bending	0.27 OK
73	38–22	Diagonal	SHS 150x150x6	5.66	-148.05	H1-1a, axial + bending	0.33 OK
74	38–5	Diagonal	SHS 150x150x6	5.66	+148.48	H1-1a, axial + bending	0.21 OK
75	39–23	Diagonal	SHS 150x150x6	5.66	-105.86	H1-1a, axial + bending	0.24 OK
76	39–6	Diagonal	SHS 150x150x6	5.66	+106.31	normal stress (H.3.4)	0.15 OK
77	40–24	Diagonal	SHS 150x150x6	5.66	-63.10	inelastic buckling (E3-2)	0.14 OK
78	40–7	Diagonal	SHS 150x150x6	5.66	+63.57	normal stress (H.3.4)	0.09 OK
79	41–25	Diagonal	SHS 150x150x6	5.66	-22.84	inelastic buckling (E3-2)	0.05 OK
80	41–8	Diagonal	SHS 150x150x6	5.66	+23.32	normal stress (H.3.4)	0.03 OK
81	42–25	Diagonal	SHS 150x150x6	5.66	-22.84	inelastic buckling (E3-2)	0.05 OK
82	42–8	Diagonal	SHS 150x150x6	5.66	+23.32	normal stress (H.3.4)	0.03 OK
83	43–26	Diagonal	SHS 150x150x6	5.66	-63.10	inelastic buckling (E3-2)	0.14 OK
84	43–9	Diagonal	SHS 150x150x6	5.66	+63.57	normal stress (H.3.4)	0.09 OK
85	44–27	Diagonal	SHS 150x150x6	5.66	-105.86	H1-1a, axial + bending	0.24 OK
86	44–10	Diagonal	SHS 150x150x6	5.66	+106.31	normal stress (H.3.4)	0.15 OK
87	45–28	Diagonal	SHS 150x150x6	5.66	-148.05	H1-1a, axial + bending	0.33 OK
88	45–11	Diagonal	SHS 150x150x6	5.66	+148.48	H1-1a, axial + bending	0.21 OK
89	46–29	Diagonal	SHS 150x150x6	5.66	-190.35	H1-1a, axial + bending	0.42 OK
90	46–12	Diagonal	SHS 150x150x6	5.66	+190.74	H1-1a, axial + bending	0.27 OK
91	47–30	Diagonal	SHS 150x150x6	5.66	-232.71	H1-1a, axial + bending	0.52 OK
92	47–13	Diagonal	SHS 150x150x6	5.66	+233.09	H1-1a, axial + bending	0.33 OK
93	48–31	Diagonal	SHS 150x150x6	5.66	-274.65	H1-1a, axial + bending	0.62 OK
94	48–14	Diagonal	SHS 150x150x6	5.66	+274.86	H1-1a, axial + bending	0.39 OK
95	49–32	Diagonal	SHS 150x150x6	5.66	-316.46	H1-1a, axial + bending	0.70 OK
96	49–15	Diagonal	SHS 150x150x6	5.66	+316.03	H1-1a, axial + bending	0.45 OK

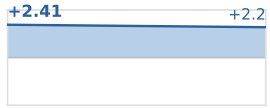
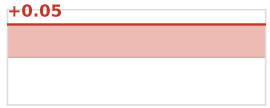
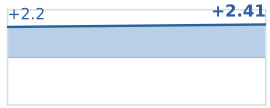
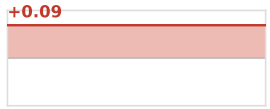
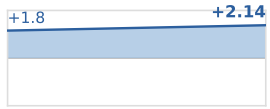
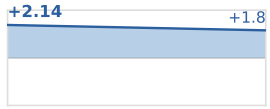
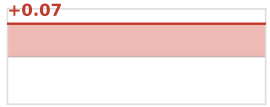
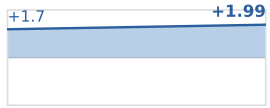
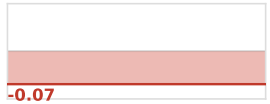
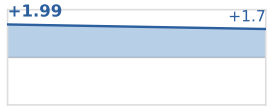
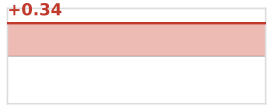
One row per rod that bends, worst first, down the left column and then the right. The thumbnail shows where the rod is; each diagram is scaled to its own rod, as on screen, with its largest value written at the peak. Rods with no shear or moment are in the Rods table.

	Where	Shear V (kN)	Moment M (kN·m)
Rod 0 use 0.05 0–1 · 4.00 m HEA 220 N +1.38 kN			
Rod 32 use 0.45 0–34 · 4.00 m SHS 200x200x8 N -448.07 kN			
Rod 63 use 0.45 16–49 · 4.00 m SHS 200x200x8 N -448.07 kN			
Rod 15 use 0.05 15–16 · 4.00 m HEA 220 N +1.38 kN			
Rod 24 use 0.67 25–26 · 4.00 m HEB 220 N -944.03 kN			
Rod 23 use 0.67 24–25 · 4.00 m HEB 220 N -944.03 kN			
Rod 22 use 0.64 23–24 · 4.00 m HEB 220 N -899.44 kN			
Rod 25 use 0.64 26–27 · 4.00 m HEB 220 N -899.44 kN			

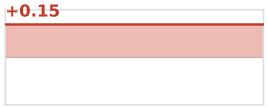
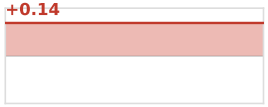
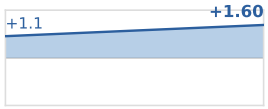
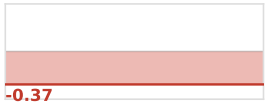
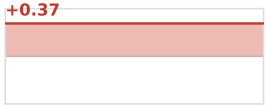
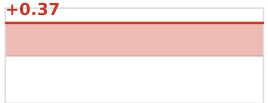
	Where	Shear V (kN)	Moment M (kN·m)
Rod 31 use 0.02 32–33 · 4.00 m HEB 220 N -1.08 kN			
Rod 33 use 0.05 34–17 · 4.00 m SHS 200x200x8 N -0.82 kN			
Rod 16 use 0.02 17–18 · 4.00 m HEB 220 N -1.08 kN			
Rod 64 use 0.05 49–33 · 4.00 m SHS 200x200x8 N -0.82 kN			
Rod 30 use 0.15 31–32 · 4.00 m HEB 220 N -225.22 kN			
Rod 17 use 0.15 18–19 · 4.00 m HEB 220 N -225.22 kN			
Rod 8 use 0.75 8–9 · 4.00 m HEA 220 N +943.77 kN			
Rod 7 use 0.75 7–8 · 4.00 m HEA 220 N +943.77 kN			

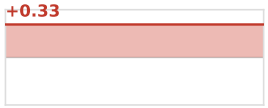
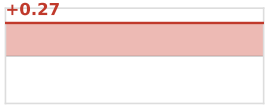
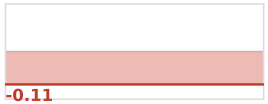
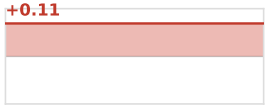
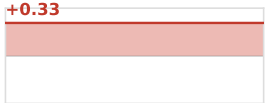
One row per rod that bends, worst first, down the left column and then the right. The thumbnail shows where the rod is; each diagram is scaled to its own rod, as on screen, with its largest value written at the peak. Rods with no shear or moment are in the Rods table.

	Where	Shear V (kN)	Moment M (kN·m)
Rod 27 use 0.51 28–29 · 4.00 m HEB 220 N -719.61 kN			
Rod 20 use 0.51 21–22 · 4.00 m HEB 220 N -719.61 kN			
Rod 19 use 0.42 20–21 · 4.00 m HEB 220 N -584.75 kN			
Rod 28 use 0.42 29–30 · 4.00 m HEB 220 N -584.75 kN			
Rod 26 use 0.58 27–28 · 4.00 m HEB 220 N -824.47 kN			
Rod 21 use 0.58 22–23 · 4.00 m HEB 220 N -824.47 kN			
Rod 18 use 0.31 19–20 · 4.00 m HEB 220 N -419.88 kN			
Rod 29 use 0.31 30–31 · 4.00 m HEB 220 N -419.88 kN			

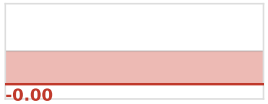
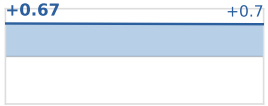
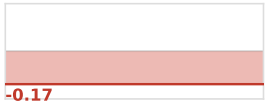
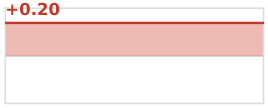
	Where	Shear V (kN)	Moment M (kN·m)
Rod 9 use 0.71 9–10 · 4.00 m HEA 220 N +899.18 kN			
Rod 6 use 0.71 6–7 · 4.00 m HEA 220 N +899.18 kN			
Rod 10 use 0.65 10–11 · 4.00 m HEA 220 N +824.23 kN			
Rod 5 use 0.65 5–6 · 4.00 m HEA 220 N +824.23 kN			
Rod 4 use 0.57 4–5 · 4.00 m HEA 220 N +719.38 kN			
Rod 11 use 0.57 11–12 · 4.00 m HEA 220 N +719.38 kN			
Rod 2 use 0.34 2–3 · 4.00 m HEA 220 N +419.68 kN			
Rod 13 use 0.34 13–14 · 4.00 m HEA 220 N +419.68 kN			

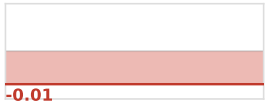
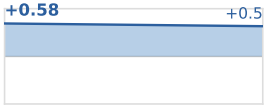
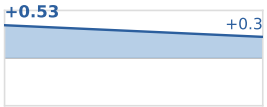
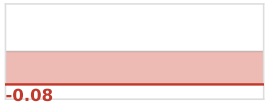
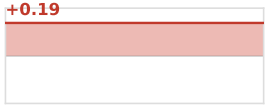
One row per rod that bends, worst first, down the left column and then the right. The thumbnail shows where the rod is; each diagram is scaled to its own rod, as on screen, with its largest value written at the peak. Rods with no shear or moment are in the Rods table.

	Where	Shear V (kN)	Moment M (kN·m)
Rod 12 use 0.47 12–13 · 4.00 m HEA 220 N +584.55 kN			
Rod 3 use 0.47 3–4 · 4.00 m HEA 220 N +584.55 kN			
Rod 14 use 0.17 14–15 · 4.00 m HEA 220 N +225.18 kN			
Rod 1 use 0.17 1–2 · 4.00 m HEA 220 N +225.18 kN			
Rod 35 use 0.51 35–18 · 4.00 m SHS 120x120x5 N +223.25 kN			
Rod 62 use 0.51 48–32 · 4.00 m SHS 120x120x5 N +223.25 kN			
Rod 37 use 0.45 36–19 · 4.00 m SHS 120x120x5 N +195.15 kN			
Rod 60 use 0.45 47–31 · 4.00 m SHS 120x120x5 N +195.15 kN			

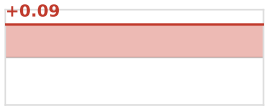
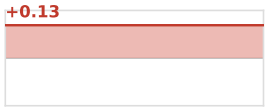
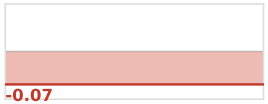
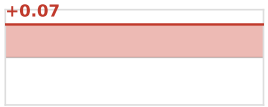
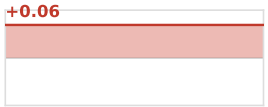
	Where	Shear V (kN)	Moment M (kN·m)
Rod 34 use 0.54 1–35 · 4.00 m SHS 120x120x5 N -165.26 kN			
Rod 61 use 0.54 15–48 · 4.00 m SHS 120x120x5 N -165.26 kN			
Rod 39 use 0.38 37–20 · 4.00 m SHS 120x120x5 N +164.41 kN			
Rod 58 use 0.38 46–30 · 4.00 m SHS 120x120x5 N +164.41 kN			
Rod 67 use 0.62 35–19 · 5.66 m SHS 150x150x6 N -274.65 kN			
Rod 93 use 0.62 48–31 · 5.66 m SHS 150x150x6 N -274.65 kN			
Rod 49 use 0.05 9–42 · 4.00 m SHS 120x120x5 N +13.88 kN			
Rod 46 use 0.05 7–41 · 4.00 m SHS 120x120x5 N +13.88 kN			

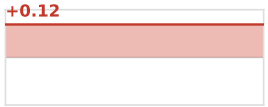
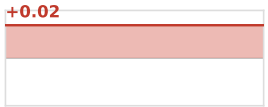
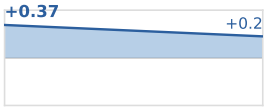
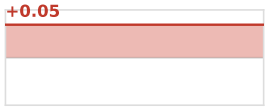
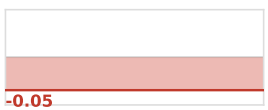
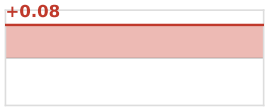
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	Where	Shear V (kN)	Moment M (kN·m)
Rod 66 use 0.45 34–1 · 5.66 m SHS 150x150x6 N +316.03 kN			
Rod 96 use 0.45 49–15 · 5.66 m SHS 150x150x6 N +316.03 kN			
Rod 36 use 0.43 2–36 · 4.00 m SHS 120x120x5 N -134.18 kN			
Rod 59 use 0.43 14–47 · 4.00 m SHS 120x120x5 N -134.18 kN			
Rod 81 use 0.05 42–25 · 5.66 m SHS 150x150x6 N -22.84 kN			
Rod 79 use 0.05 41–25 · 5.66 m SHS 150x150x6 N -22.84 kN			
Rod 41 use 0.31 38–21 · 4.00 m SHS 120x120x5 N +134.67 kN			
Rod 56 use 0.31 45–29 · 4.00 m SHS 120x120x5 N +134.67 kN			

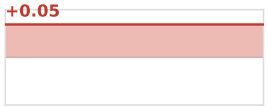
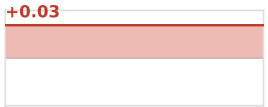
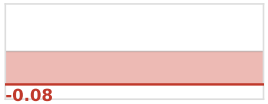
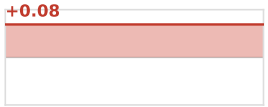
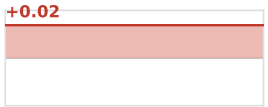
	Where	Shear V (kN)	Moment M (kN·m)
Rod 65 use 0.70 34–18 · 5.66 m SHS 150x150x6 N -316.46 kN			
Rod 95 use 0.70 49–32 · 5.66 m SHS 150x150x6 N -316.46 kN			
Rod 94 use 0.39 48–14 · 5.66 m SHS 150x150x6 N +274.86 kN			
Rod 68 use 0.39 35–2 · 5.66 m SHS 150x150x6 N +274.86 kN			
Rod 69 use 0.52 36–20 · 5.66 m SHS 150x150x6 N -232.71 kN			
Rod 91 use 0.52 47–30 · 5.66 m SHS 150x150x6 N -232.71 kN			
Rod 51 use 0.05 10–43 · 4.00 m SHS 120x120x5 N -15.06 kN			
Rod 44 use 0.05 6–40 · 4.00 m SHS 120x120x5 N -15.06 kN			

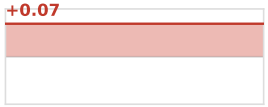
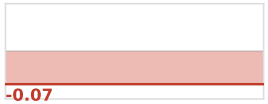
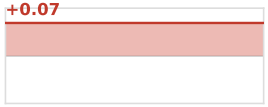
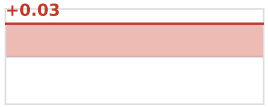
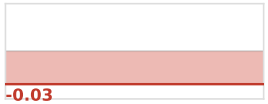
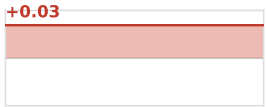
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	Where	Shear V (kN)	Moment M (kN·m)
Rod 71 use 0.42 37–21 · 5.66 m SHS 150x150x6 N -190.35 kN			
Rod 89 use 0.42 46–29 · 5.66 m SHS 150x150x6 N -190.35 kN			
Rod 42 use 0.13 5–39 · 4.00 m SHS 120x120x5 N -45.17 kN			
Rod 53 use 0.13 11–44 · 4.00 m SHS 120x120x5 N -45.17 kN			
Rod 38 use 0.33 3–37 · 4.00 m SHS 120x120x5 N -105.02 kN			
Rod 57 use 0.33 13–46 · 4.00 m SHS 120x120x5 N -105.02 kN			
Rod 84 use 0.09 43–9 · 5.66 m SHS 150x150x6 N +63.57 kN			
Rod 78 use 0.09 40–7 · 5.66 m SHS 150x150x6 N +63.57 kN			

	Where	Shear V (kN)	Moment M (kN·m)
Rod 43 use 0.23 39–22 · 4.00 m SHS 120x120x5 N +104.81 kN			
Rod 54 use 0.23 44–28 · 4.00 m SHS 120x120x5 N +104.81 kN			
Rod 70 use 0.33 36–3 · 5.66 m SHS 150x150x6 N +233.09 kN			
Rod 92 use 0.33 47–13 · 5.66 m SHS 150x150x6 N +233.09 kN			
Rod 83 use 0.14 43–26 · 5.66 m SHS 150x150x6 N -63.10 kN			
Rod 77 use 0.14 40–24 · 5.66 m SHS 150x150x6 N -63.10 kN			
Rod 75 use 0.24 39–23 · 5.66 m SHS 150x150x6 N -105.86 kN			
Rod 85 use 0.24 44–27 · 5.66 m SHS 150x150x6 N -105.86 kN			

One row per rod that bends, worst first, down the left column and then the right. The thumbnail shows where the rod is; each diagram is scaled to its own rod, as on screen, with its largest value written at the peak. Rods with no shear or moment are in the Rods table.

	Where	Shear V (kN)	Moment M (kN·m)
Rod 80 use 0.03 41–8 · 5.66 m SHS 150x150x6 N +23.32 kN			
Rod 82 use 0.03 42–8 · 5.66 m SHS 150x150x6 N +23.32 kN			
Rod 40 use 0.23 4–38 · 4.00 m SHS 120x120x5 N -74.98 kN			
Rod 55 use 0.23 12–45 · 4.00 m SHS 120x120x5 N -74.98 kN			
Rod 73 use 0.33 38–22 · 5.66 m SHS 150x150x6 N -148.05 kN			
Rod 87 use 0.33 45–28 · 5.66 m SHS 150x150x6 N -148.05 kN			
Rod 72 use 0.27 37–4 · 5.66 m SHS 150x150x6 N +190.74 kN			
Rod 90 use 0.27 46–12 · 5.66 m SHS 150x150x6 N +190.74 kN			

	Where	Shear V (kN)	Moment M (kN·m)
Rod 52 use 0.15 43–27 · 4.00 m SHS 120x120x5 N +74.52 kN			
Rod 45 use 0.15 40–23 · 4.00 m SHS 120x120x5 N +74.52 kN			
Rod 50 use 0.10 42–26 · 4.00 m SHS 120x120x5 N +46.36 kN			
Rod 47 use 0.10 41–24 · 4.00 m SHS 120x120x5 N +46.36 kN			
Rod 76 use 0.15 39–6 · 5.66 m SHS 150x150x6 N +106.31 kN			
Rod 86 use 0.15 44–10 · 5.66 m SHS 150x150x6 N +106.31 kN			
Rod 74 use 0.21 38–5 · 5.66 m SHS 150x150x6 N +148.48 kN			
Rod 88 use 0.21 45–11 · 5.66 m SHS 150x150x6 N +148.48 kN			



Rod 8

utilisation 0.75

H1-1a, axial + bending

```
Rod 8
=====
Steel, CIRSOC 301-2018 -- A = 6,156 mm², Fy = 235 MPa, L = 4.000 m

 1. Axial stress
      σ = N / A
      = 943,766 N / 6,156 mm²
      = 153.31 MPa

 2. Design stress  [H.3.4]
      φ · Fy
      = 0.9 × 235
      = 211.50 MPa

 3. Tension ratio  [H.3.4]
      σ / (φ Fy)
      = 153.31 / 211.50
      = 0.725

 4. Bending capacity  [F.1]
      Mc = φb · Fy · S
      = 0.9 × 235 × 493,734 mm³
      = 104.425 kN·m

 5. Axial + bending  [H1-1a]
      Pr/Pc + 8/9 (Mry/Mc + Mrx/Mcw)
      = 0.725 + 8/9 (0.000/104.425 + 0.000/104.425)
      = 0.754

 6. Governs
      the largest ratio
      = H1-1a, axial + bending
      = 0.754

utilisation 0.754 -- OK
```



Rod 7

utilisation 0.75

H1-1a, axial + bending

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Rod 7
=====
Steel, CIRSOC 301-2018 -- A = 6,156 mm², Fy = 235 MPa, L = 4.000 m

 1. Axial stress
      σ = N / A
      = 943,766 N / 6,156 mm²
      = 153.31 MPa

 2. Design stress  [H.3.4]
      φ · Fy
      = 0.9 × 235
      = 211.50 MPa

 3. Tension ratio  [H.3.4]
      σ / (φ Fy)
      = 153.31 / 211.50
      = 0.725

 4. Bending capacity  [F.1]
      Mc = φb · Fy · S
      = 0.9 × 235 × 493,734 mm³
      = 104.425 kN·m

 5. Axial + bending  [H1-1a]
      Pr/Pc + 8/9 (Mry/Mc + Mrx/Mcw)
      = 0.725 + 8/9 (0.000/104.425 + 0.000/104.425)
      = 0.754

 6. Governs
      the largest ratio
      = H1-1a, axial + bending
      = 0.754

utilisation 0.754 -- OK
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